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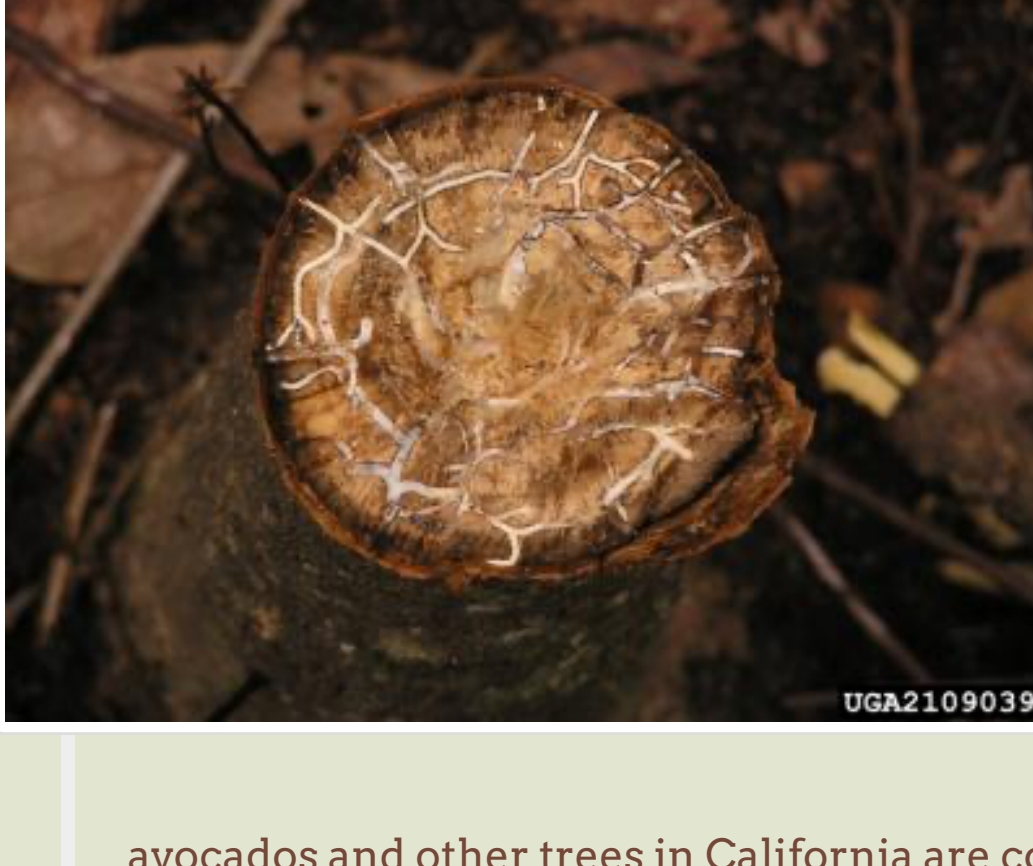
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LAUREL WILT

*Harringtonia lauricola* & *Xyleborus glabratus**Eichhoff***Last updated by:**
Faith Campbell*August 2021*

Laurel wilt – redbay ambrosia beetle *Xyleborus glabratus* and associated fungus *Harringtonia lauricola*

NOTE: the ambrosia beetle and fungus complex attacking

avocados and other trees in California are completely unrelated organisms to Laurel wilt, despite both being beetle-fungi complexes that affect avocado trees. Click [here to read about the polyphagous shot hole borer and *Fusarium* fungus](#).

USDA-1090339

Beginning in 2003, scientists began receiving reports of dying redbay trees (*Persea borbonia*) in coastal areas of Georgia and South Carolina (J. Johnson, Georgia Forestry Commission, pers. comm. 4 April 2005), then in northeast Florida (J. Foltz, University of Florida, pers. comm. November 2005). The cause was determined to be a previously unknown fungus, *Harringtonia lauricola*, formerly *Raffaelea lauricola*) transported (vectored) by a recently introduced ambrosia beetle from Asia, *Xyleborus glabratus* (Fraedrich *et al.* 2008). The beetle had first been detected in the United States in 2002 near Savannah, Georgia (Rabaglia *et al.* 2006).

The disease has since spread rapidly. It now is found in 11 states, reaching from eastern North Carolina south along most of the Florida peninsula; and west to eastern Texas; north to Kentucky (Olatinwo *et al.* 2021). One infected Tennessee county is on the border with Virginia. The most recent Laurel wilt map by USA Counties is available here: <https://www.arcgis.com/apps/dashboards/443391c8fdb741b597e6ccf1236d2a02>

The beetle can apparently fly several kilometers; it might also be carried on wind currents (Mayfield *et al.* 2009). There is also evidence that it can be transported by human movement of infested wood. Thus, in addition to the initial detection of the beetle near a port which receives significant amounts of wood packaging, several isolated outbreaks can be traced to movement of wood – e.g., those near a hardwood mulch plant and a state park in Georgia (Cameron *et al.* 2008); and near the shop of a woodturner who collected redbay logs from an infested area in Florida (Mayfield *et al.* 2009). The suspected pathways of long distance spread include infested solid wood packing material, firewood, and logs (Mayfield *et al.* 2009). Wood chips are no longer considered likely to spread either the beetle or the fungus (J. Smith, 2015).

Because the beetle can reproduce without mating, a population in a new, isolated location can potentially be started by only one female – as long as it finds suitable host material (Mayfield *et al.* 2009).

After a redbay tree has died, it is typically attacked by additional redbay ambrosia beetles. Later, female beetles emerging from the dead redbay tree disperse in search of new hosts. Dead redbay trees can serve as host material for the beetle for many weeks after initial colonization. Many coastal forests have up to hundreds of redbay trees per acre, allowing redbay ambrosia beetle populations to build rapidly (Mayfield *et al.* 2009).

The redbay ambrosia beetle is native to Asia where its reported hosts include several species in the families Lauraceae, Dipterocarpaceae, Fagaceae (oak family), and Fabaceae (bean family) (Rabaglia *et al.* 2006). No laurel wilt disease is reported in these or other plant species in the beetle's native range.

Redbay ambrosia beetle is the principal vector of the pathogen (Mayfield *et al.* 2009). However, it is now suspected that other ambrosia beetle species can transfer the pathogen. This appears to be occurring especially in avocado, which is rarely attacked by the redbay ambrosia beetle (T. Dreaden and D. Carrillo, 2015). Attack by even a single female beetle is sufficient to inoculate the tree with sufficient numbers of spores of *Harringtonia lauricola* to initiate disease (Mayfield *et al.* 2009). The pathogen moves throughout the tree through the sapwood, presumably causing a restriction in the flow of water and wilting of the leaves. In redbay, the entire crown eventually wilts and turns brown over a period that may take from a few weeks to 2-3 months (Mayfield *et al.* 2009).

More than 90% of redbays with stems greater than 1 inch diameter have died within 2 years in some stands (Fraedrich *et al.* 2008). Smaller diameter redbay seedlings and sprouts appear to be much less frequently affected by the disease in the field (Fraedrich *et al.* 2008). By 2017, an estimated 320 million trees – nearly one-third of all redbays – had been killed (Hughes *et al.* 2017).

Scientists expect continued dramatic reductions in redbay populations, although survival of redbay regeneration in the aftermath of laurel wilt epidemics suggests that redbay will not go extinct (Mayfield *et al.* 2009). A team of USDA Forest Service scientists has predicted that redbay will lose 90% of its basal area before 2030 (USDA FS 2014). Redbay was ranked by the IUCN in 2018 as “least concern”, but as of the writing of the Morton report (Beckman *et al.* 2021), it was being re-assessed by the IUCN with the expectation its ranking would change to “vulnerable”.

While regeneration from seed and root sprouts is robust, scientists expect these trees to succumb once they reach sufficient size to be attacked by the redbay ambrosia beetle (Hughes *et al.* 2017).

Laurel wilt has dramatically altered forest composition and dynamics in the coastal forests of the southeast – which are considered a biological diversity “hot spot” (Hughes *et al.* 2017).

Redbay is important to wildlife. The fruit is eaten by wild turkey, bobwhite quail and several species of song birds. Deer also consume the fruits and leaves (Brendemuehl, 1990). An additional ecological concern is the dependence of the Palamedes swallowtail butterfly (*Papilio palamedes*) on redbay and swamp bay (*Persea palustris*); the butterfly's larvae feed on these trees – both of which have been killed in large numbers (Mayfield *et al.* 2013). The adult Palamedes swallowtail butterfly is the primary pollinator of yellow-fringed orchid (J. Riggins 2015). A second butterfly, the spicebush swallowtail (*Papilio troilus*), is also dependent on species in the family Lauraceae and might be harmed if the pathogen proves to damage additional woody plants in that family (J. Foltz, University of Florida, pers. comm. November 2005).

Several other tree and shrub species in the Lauraceae family are also at risk. Swamp bays (*Persea borbonia* var. *pubescens* or *P. palustris*) grow in moist woodlands, savannas, swamps, and wetlands in the Atlantic and Gulf Coast Plain, and the Piedmont, from Pennsylvania to east Texas (Beckman *et al.* 2021). In the everglades ecosystem, swamp bay is especially on “tree islands” scattered through the sawgrass swamp. Tree islands comprise only about 10% of the everglades landcover, but contribute greatly to biological diversity and ecosystem process, including as wet season refugia for mammals. The result of widespread swamp bay mortality is expected to be reduced food availability for dependent mammals and insects and – where the canopy is opened, invasion by non-native plants (Pernas, 2015).

Laurel wilt has spread quickly across Everglades National Park and Big Cypress National Preserve; the extent of infected trees covered 372,000 ha in just four years from the first detection in 2011. The disease has reached the edge of vulnerable trees on the Southwest side, but can still spread to the North. Officials believe the rapid spread was helped by the wind, which blows predominantly East to West from February through October. The disease has not become virulent in the Loxahatchee National Wildlife Refuge to the Northeast; the reason is unclear but might be related to the fact that this area does not have a prolonged dry spell (Rodgers, 2015). At the time of writing (Beckman *et al.* 2021), swamp bay was under assessment by the IUCN; it was recommended for the “vulnerable” category.

Redbay and swamp bay provide key ingredients in 90% of traditional medicines used by the Seminole & Miccosukee tribes of Florida. The tribes are supporting scientific efforts to find, propagate, and restore swamp bay trees resistant to the disease (Sobel, 2015; Smith, 2015).

Laurel wilt infection of sassafras (*Sassafras albidum*) has expanded in recent years. Local spread might be primarily through root grafts. However, the many “jumps” to disjunct areas where sassafras occurs in isolation from other hosts is probably a result of human or wind transport (Olatinwo *et al.* 2021).

Sassafras is widely distributed throughout the eastern United States on dry ridges and upper slopes and as a pioneer species. An estimated 1.9 billion sassafras trees are found across 28 states – reaching to southern Michigan. Sassafras grows in 53 ecoregions and 69 forest types. While currently 52% of the range of sassafras might experience winter temperatures sufficiently cold to cause significant mortality of the primary vector beetle, a modest warming of the climate (1.4 °C increase in winter minimum temperatures by 2050) would reduce the protected area to less than ten percent of the current sassafras range (Olatinwo *et al.* 2021). Because of sassafras' large range, in 2020 the IUCN ranked it as of “least concern” (Beckman *et al.* 2021).

Silk bay (*Persea humilis*) is endemic to FL and occurs in sand pine-scrub and shrub layers dominated by evergreen oaks. This species' status is being assessed by the IUCN; it is recommended for the “near threatened” category (Beckman *et al.* 2021).

Several shrub species appear to be vulnerable laurel wilt disease but are protected by the redbay ambrosia beetle's preference for stems of larger-diameter stems. These include two rare species – pondspice (*Litsea aestivalis*) (listed as endangered in South Carolina but found through much of the southeastern coastal plain) and the federally listed pondberry (*Lindera melissifolia*). The northern spicebush (*Lindera benzoin* (L.) Blume) enjoys similar protection (Olatinwo *et al.* 2021). As of 2020, the IUCN Red List considered *Litsea aestivalis* and *L. benzoin* to be of “least concern” (Beckman *et al.* 2021).

An important tree or shrub of the far West – California laurel (*Umbellularia californica*) – appears to be vulnerable (Olatinwo *et al.* 2021). This evergreen grows across a variety of sites in the Pacific coastal regions of Southwest Oregon, western California, and the foothills of the Cascade Range and Sierra Nevada mountains. The fruit is fed on by squirrels, jays and other animals.

Like redbay, California bay laurel has attractive wood used for cabinetry, furniture, veneer and other specialty wood products (Mayfield *et al.* 2013). This means that there is a risk that the disease could be transported by the woodworker community. A complicating factor is that California bay laurel is a reproductive host for the sudden oak death (SOD) pathogen, *Phytophthora ramorum*. Bay laurels growing in SOD-infested counties are subject to quarantine restrictions.

The greatest economic impact from laurel wilt disease could be to Florida avocados (*Persea americana* Mill.); yard trees and experimentally planted avocados in coastal Florida have been killed (Mayfield *et al.* 2009). The disease was detected in commercial orchards in 2012 (Carrillo, 2015). The avocado industry has prepared a recovery plan. The plan calls for early detection and rapid removal of diseased trees to avoid creating “beetle factories”. As of June 2015, 9,000 trees had been removed – about 1% of total numbers. This removal is expensive because of the size of the trees and their interlocking roots – so some landowners have abandoned their orchards. Since the disease is not considered a quarantine pest, there are no legal requirements that growers follow the protocol (Carrillo, 2015).

Management of the disease in avocado groves is complicated by the variety of ways in which the disease can spread – both by several of the ambrosia beetle species common in the area and by neighboring trees' feeding their root systems. The avocado industry is funding intensive insect trapping and research into control methods (Carrillo, 2015).

It appears to be too late to stop the spread of laurel wilt throughout the range of redbay and perhaps into other plant communities as well. Research is needed on tools to manage the beetle and the disease. One strategy is to breed trees that are resistant to or tolerant of the disease (https://foresthealth/doc/ForestHealth_Description.pdf)

A USDA Forest Service research project, “Project CAPTURE” [<https://forestthreats.org/research/projects/project-summaries/genetic-risk-assessment-system>] (Conservation Assessment and Prioritization of Forest Trees Under Risk of Extirpation) established priorities for coordinated conservation efforts, including resistance breeding. Redbay was placed in the top priority for such efforts.

A partnership led by the Morton Arboretum has prepared an assessment of the threat from laurel wilt to Lauraceae species native to the United States. The assessment's species-specific conclusions are reported here. The assessment provides more complete descriptions of the species' ranges and considers *ex situ* conservation efforts (Beckman *et al.* 2021).

California avocado growers are dealing with a separate insect/pathogen disease complex. The **polyphagous shot hole borer** is a previously undescribed beetle in the *Euwallacea* genus that vectors a *Fusarium* fungus – also previously undescribed. The disease complex is already established in the Los Angeles area.

Laurel wilt appears to threaten trees in the Lauraceae family in other regions, too. Central and South America are home to more than 750 lauraceae species in 26 genera (Olatinwo *et al.* 2021). The presence of the disease in eastern Texas raises considerable concern. Numerous Lauraceae also grow in Puerto Rico (D.J. Lodge, U.S. Forest Service, pers. comm. June 2013)

Other forest systems with numerous apparently vulnerable hosts include Australia, with 125 native lauraceous species in eight genera; Madagascar, with 135 species in six genera; and the

Laurasilva forests on the Macaronesian Islands off the coasts of Europe and Africa (the Azores, Canary Islands, and Madeira). The commercial spice bay laurel (*Laurus nobilis*) is native to mainland Europe. Perhaps its small stem diameter and discontinuous distribution and the absence of other lauraceous host species reduce the probability of a widespread epidemic in this last species (Olatinwo *et al.* 2021).

A compelling article about laurel wilt, as it is killing redbay (*Persea borbonia*), was written by Susan Cerulean, and has been posted on line at <https://terrains.org/articles/22/cerulean.htm>.

For more information on this pest/fungal complex, please visit:

- Florida Division of Plant Industry, Laurel Wilt information
- <https://southernforesthealth.net/diseases/laurel-wilt/distribution-map>

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Ohi'a (*Austropuccinia*) Rust

Pine Needle Scale

